

Material

Formelsammlung

- $\mathbb{N} = \mathbb{N} \setminus \{0\} = \{1, 2, 3, \dots\}$
 $\mathbb{N}_0 = \{0, 1, 2, \dots\}$
- Exponentialfunktion:
 $a^b a^c = a^{b+c}$
 $a^c b^c = (ab)^c$
 $a^{b^c} = a^{(b^c)}$
 $(a^b)^c = a^{bc}$
 $a^{1/b} = \sqrt[b]{a}$
 $a^b = e^{b \ln a}$
- Logarithmus:
 $\log a + \log b = \log(a \cdot b)$
 $\log a^b = b \cdot \log a$
 $\log_a n = \log_a b \cdot \log_b n$
 $a^{\log b} = b^{\log a}$
 $\log 1 = 0$
- $\mathcal{O}$ -Notation: $g \in \mathcal{O}(f) \iff \exists C > 0 : \forall n \geq n_0 : g(n) \leq C \cdot f(n)$.
- Ω -Notation: $g \in \Omega(f) \iff f \in \mathcal{O}(g)$.
- Θ -Notation: $g \in \Theta(f) \iff f \in \mathcal{O}(g) \wedge g \in \mathcal{O}(f)$.
- Asymptotic growth: $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \begin{cases} 0 : & f \in \mathcal{O}(g) \\ C : & f \in \Theta(g) \\ \infty : & f \in \Omega(g) \end{cases}$
- L'Hôpital: $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)}$, if $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} g(x) = 0$ or $\pm\infty$.
- Master theorem: $T(n) \leq aT(n/2) + Cn^b : \begin{cases} T(n) \leq \mathcal{O}(n^b), & b > \log_2 a \\ T(n) \leq \mathcal{O}(n^b \log n), & b = \log_2 a \\ T(n) \leq \mathcal{O}(n^{\log_2 a}), & b < \log_2 a. \end{cases}$
If T is increasing, this holds $\forall n$ (else only for $n = 2^k$).
If above equation holds with "=", $\mathcal{O}$ can be replaced with Θ .

Algorithmen

GROUP	ALGORITHM	RUNTIME
	Karatsuba	$\mathcal{O}(n^{\log_2(3)})$
Search	Binary Search	$\mathcal{O}(\log n)$
	Linear Search	$\mathcal{O}(n)$

GROUP	ALGORITHM	RUNTIME
Sort	IsSorted	$\mathcal{O}(n)$
	BubbleSort	$\mathcal{O}(n^2)$
	SelectionSort	$\mathcal{O}(n^2)$
	InsertionSort	$\mathcal{O}(n^2)$
	Heapsort	$\mathcal{O}(n \log n)$
	Mergesort	$\mathcal{O}(n \log n)$
	Quicksort	ac: $\mathcal{O}(n \log n)$ (wc: $\mathcal{O}(n^2)$)
	Quicksort groups of 5	$\mathcal{O}(n \log n)$
	Radixsort	$\mathcal{O}(n)$
DP	Max subarray sum (contiguous)	$\mathcal{O}(n)$
	Longest increasing subarray (non-cont.)	$\mathcal{O}(n \log n)$
	Longest common substring (non-cont.)	$\mathcal{O}(nm)$
	Minimal editing distance	$\mathcal{O}(nm)$
	Subset sum	$\mathcal{O}(2^n n)$ or $\mathcal{O}(nb)$
	Knapsack problem	$\mathcal{O}(nW)$ or $\mathcal{O}(nV)$
	Knapsack approximative	$\mathcal{O}(\frac{n^2 v_{max}}{K}) = \mathcal{O}(\frac{n^3}{\epsilon})$
	Matrix chain multiplication	$\mathcal{O}(n^3)$
Data struct.	Stack	$\mathcal{O}(1)$
	Queue	$\mathcal{O}(1)$
	Priority Queue	$\mathcal{O}(\log n)$
	Dictionary (AVL tree)	$\mathcal{O}(\log n)$
Graphs	Eulerian walk	$\mathcal{O}(n + m)$
Topo. sort	DFS	$\mathcal{O}(n + m)$
Topo. sort	BFS	$\mathcal{O}(n + m)$
MST	Boruvka	$\mathcal{O}((n + m) \log n)$
MST	Prim	$\mathcal{O}((n + m) \log n)$
MST	Kruskal	$\mathcal{O}(n \log n + m \log m)$
SSSP	Dijkstra	$\mathcal{O}((n + m) \log n)$ or $\mathcal{O}(m + n \log n)$
SSSP	Bellman-Ford	$\mathcal{O}(nm)$
APSP	Floyd-Warshall	$\mathcal{O}(n^3)$
APSP	Johnson	$\mathcal{O}(mn + n^2)$

ac is average case, wc is worst case

MST is minimal spanning tree, SSSP is single source shortest paths, APSP is all pairs shortest paths